

FRAMEWORKS, THIRD-PARTY LIBRARIES AND TOOLS

- Django – <https://www.djangoproject.com/>
 - Django was chosen as the core web framework due to its rapid development capabilities, built-in security features, and modularity. The template-based frontend is a lightweight solution and SQLite integration allows for persistence for prompt and response sessions
- Django REST Framework – <https://www.django-rest-framework.org>
 - Django REST Framework is used to create a RESTful backend that handles prompt submission and response gathering. The Django REST Framework also provides serializers that transform JSON data into Python objects, GET/POST validation, and API abstraction.
- Hugging Face Inference API – <https://huggingface.co/inference-api>
 - The Hugging Face Inference API was chosen to be able to interact with multiple LLMs through a common interface. By using the Hugging Face service, the operational overhead of hosting models locally is avoided and rapid prototyping is possible. Encapsulating the API calls in a service layer also mitigates risks if there are rate limit or provider changes.
- Docker – <https://www.docker.com>
 - Docker is used to containerize the application to promote modularity, reproducibility, and consistency across operating systems. Containerization also benefits future upgrades and migration to cloud-based services or alternative database solutions.
- SQLite – <https://www.sqlite.org>
 - SQLite was chosen to act as a lightweight database for a single-node prototype. It integrates easily with Django.